BI/Analytics Engineering Project

E-Commerce Cohort Analysis Pipeline

- Executive Summary
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EXECUTIVE SUMMARY

E-Commerce Cohort Analysis Pipeline

Cohort & Retention Dynamics

Main Question: How long does it take for customers to make their second purchase, and how does this behavior vary across different monthly cohorts?

Goal: The objective of this project was to transition from static reporting to a dynamic **ELT (Extract, Load, Transform)** framework. By building an automated pipeline, we aimed to measure the velocity of repeat purchases and identify which customer classes or cohorts provide the highest long-term value as the business scales from 2024 into 2025.

The Process: We implemented a three-stage Medallion Architecture using Databricks. This process, known as ELT, allows us to load raw data first and transform it directly within the warehouse for maximum efficiency:

1. **Bronze:** Ingestion of raw transactional logs.
2. **Silver:** Cleaning and normalizing 2,000+ records to ensure "one version of the truth.
3. **Gold:** Calculation of high-level business logic, specifically isolating the time-lag between the first and second purchase for every unique customer.

The Outcome: We successfully validated the pipeline by performing an incremental update. After appending 1,000 new records from 2025, the system automatically recalculated all retention curves without manual intervention. The result is a dashboard that provides a real-time health check on customer loyalty, proving the technical infrastructure can handle continuous growth.

Predictive Outlook: Our analysis indicates Cohort Stabilization. By observing 2024 customers through the lens of 2025 data, we see that late-year cohorts are beginning to reach their second-purchase peak. This allows us to predict future revenue based on the first 30 days of a new cohort's life cycle with increasing accuracy. We are no longer guessing how 2025 will perform; we are measuring its foundation in real-time.

Strategic Recommendation: Based on the **Repeat Purchase Depth** metrics, we recommend a targeted re-engagement window. We should automate a Win-Back marketing campaign triggered exactly 5 days before the average threshold time-to-second-purchase and maximize the conversion of one-time buyers into loyalists.

TECHNICAL PIPELINE DESCRIPTION

E-Commerce Cohort Analysis Pipeline

Introduction

This project utilizes a **Medallion Architecture** to architect a robust, end-to-end data pipeline. The primary objective was to move beyond manual data preparation by establishing an automated, **ELT (Extract, Load, Transform)** workflow. This system ensures that raw transactional data is processed into refined, high-level business metrics that measure the health of our customer base in real-time.

Data Architecture

To maintain data integrity and traceability, the pipeline is organized into three distinct logical layers:

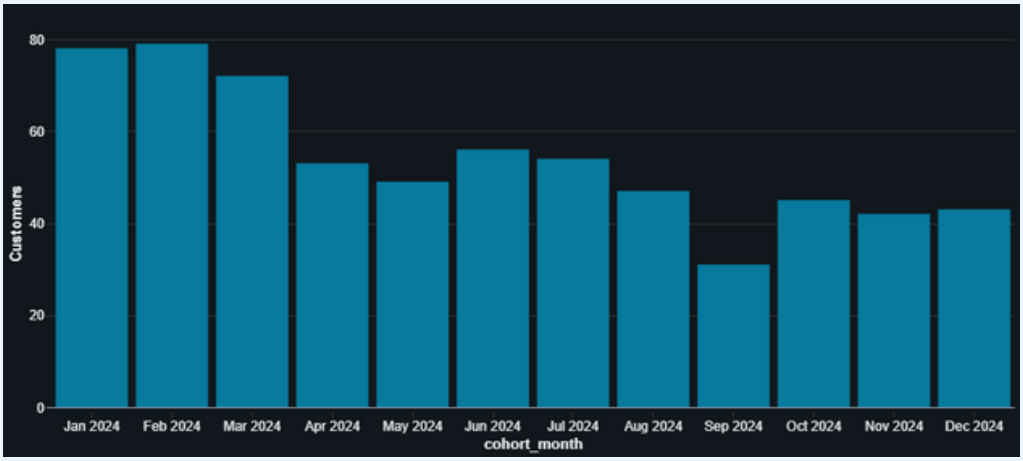
- **Bronze (Raw Ingestion):** This layer serves as the landing zone for raw CSV transaction logs. By maintaining data in its original state, we ensure a permanent audit trail for all historical 2024 and 2025 records.
- **Silver (Standardization & Validation):** In this intermediate phase, the data is "cleaned." We perform schema enforcement, standardize date formats, handle null values, and deduplicate customer records. This ensures that the downstream analysis is built upon a verified, high-quality foundation.
- **Gold (Business Logic & Aggregation):** The final layer transforms clean data into actionable insights. Here, we calculate complex cohort metrics, specifically the **time-lag between a user's first and second purchase**, and group customers into distinct monthly cohorts for trend analysis.

Scalability and Automation

The architecture leverages **Delta Lake** technology to handle incremental updates. This was demonstrated when the 2025 data was appended; the SQL logic automatically identified the new records and updated the Gold-layer metrics without any manual code changes. Because the dashboard is linked directly to these Gold tables, the transition from raw data ingestion to visual insight is fully automated, providing a **scalable solution** for future datasets.

BUSINESS VISUALIZATION OVERVIEW

Cohort Size
2024



649 total customers acquired in 2024

1. **Cohort Maturity Issue:**

- Q4 2024 cohorts (Oct-Dec) are too recent to have meaningful retention metrics
- They need 3-6 months to show true retention behavior

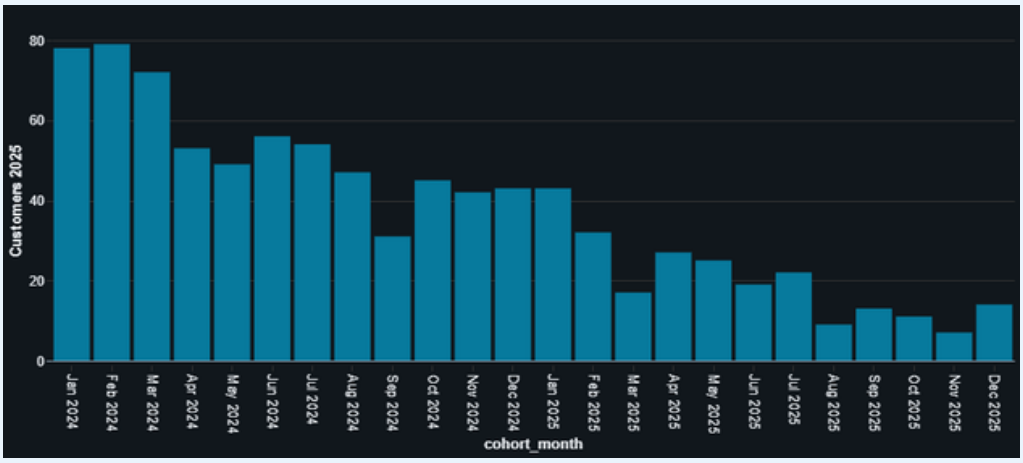
2. **Retention Variation:**

- Massive spread in days_to_second_purchase (4 to 656 days)
- Suggests different customer segments with different purchase cycles
- Need to segment by customer type for better analysis

BUSINESS VISUALIZATION OVERVIEW

Cohort Size

2024 & 2025

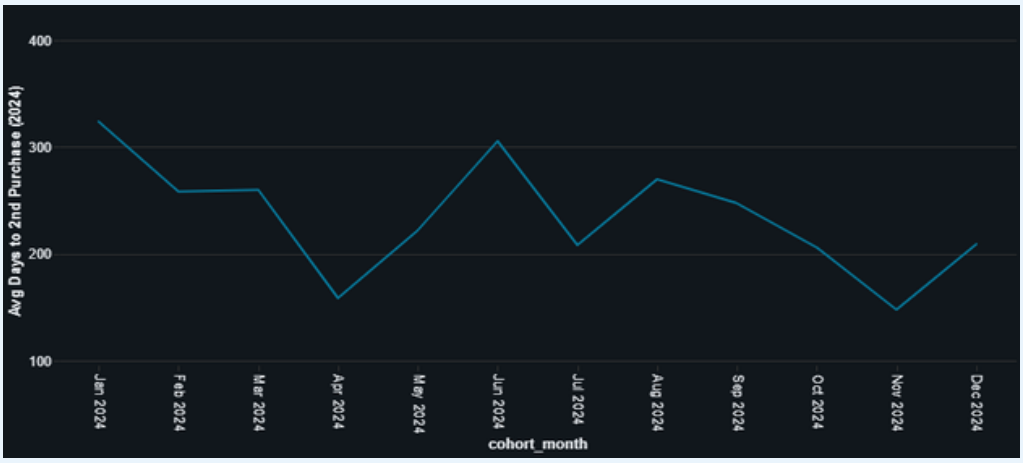


239 New Customers (+36.8% Growth)

- **Grand Total 888 customers.**
- **Extended Retention Visibility:** Many 2024 customers show second purchases.
- **Time-Series Continuity:** The business didn't stop in December 2024.
- **Complete Cohort Lifecycles:** 2024 customers can now show their long-term retention (6-12 months).
- **Year-over-Year Comparison:** You can now see if Jan 2025 acquired more/fewer customers than Jan 2024.
- **Seasonal Patterns:** With 24 months, you can identify recurring monthly/quarterly trends.

BUSINESS VISUALIZATION OVERVIEW

Retention to 2nd Purchase
2024

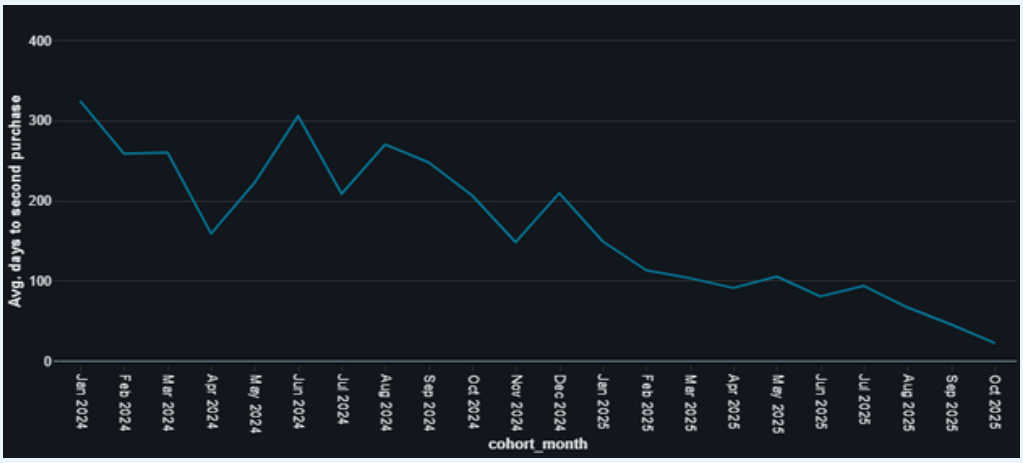


Total Repeat Customers: 427 out of 649 (65.8% overall repeat rate)

- **Incomplete Long-Term Picture:** Many early 2024 cohorts (Jan-Apr) show their second purchases in 2025.
- **Wide Retention Spectrum:** Days to second purchase ranges from 6 to 634 days.
- **Fast Returners Exist:** Several customers returned within 30 days.
- **Long Purchase Cycles:** The average is between 200-300 days.
- **Q4 2024 Bias:** Late-2024 cohorts (Oct-Dec) who are included likely only represent the fastest returners.

BUSINESS VISUALIZATION OVERVIEW

Retention to 2nd Purchase
2024 & 2025

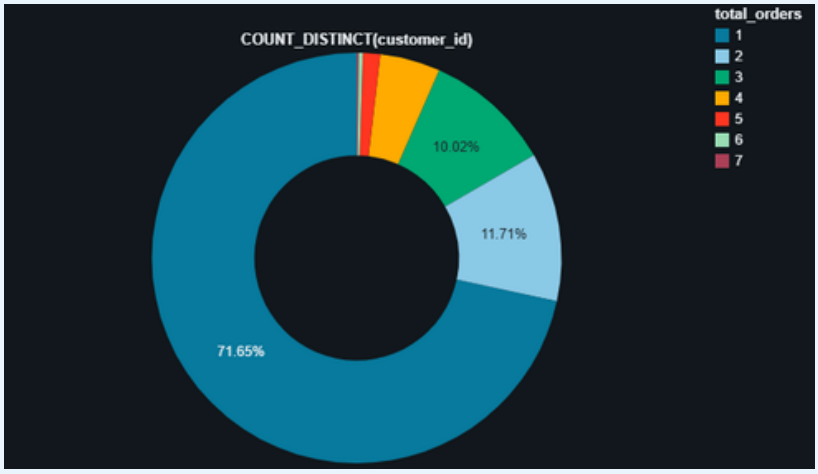


Total Repeat Customers: 580 out of 888 (65.3% overall repeat rate)

- **153 Additional Repeat Customers Revealed (+35.8% increase):** Adding 2025 data uncovered 153 customers who made their second purchase.
- **2025 Cohorts Show Faster Engagement:** New 2025 customers demonstrate notably quicker return behavior.
- **Complete Lifecycle Visibility:** Early 2024 cohorts (Jan-Mar) now show their full 12-15 month retention journey.
- **Retention Rate Stability Despite Growth:** The overall repeat rate dropped only slightly from 65.8% to 65.3% despite adding 239 new customers.
- **Actionable Segmentation Opportunity:** The combined dataset now clearly shows three distinct customer segments - Fast Returners (<90 days), Standard Returners (90-365 days), and Long-Cycle Returners (365+ days).

BUSINESS VISUALIZATION OVERVIEW

Repeat Purchase Depth
2024

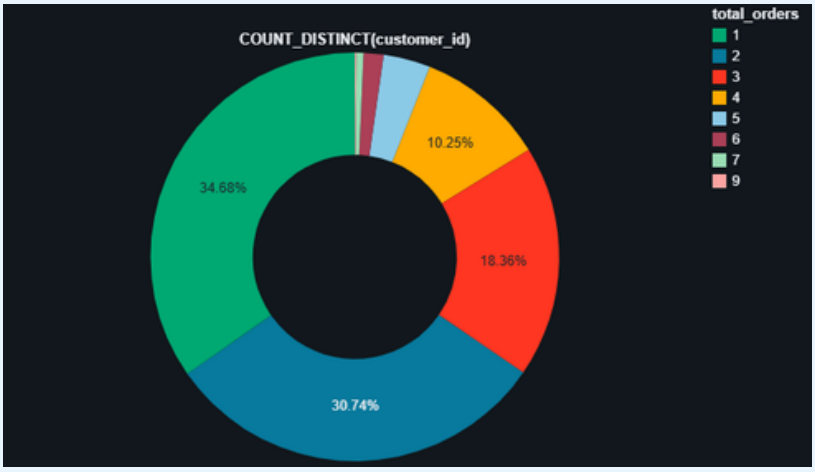


Total Customers: 649 | Distribution: Mostly single-order customers visible

- **One-Time Buyer Dominance**: The vast majority of visible customers in the 2024-only view show total_orders = 1
- **Incomplete Loyalty Picture:** Even customers who did make repeat purchases in 2024 couldn't show their 2025 orders.
- **False Churn Signal:** Many customers appearing as one-time buyers were actually in their natural purchase cycle but the data cutoff at December 2024.
- **No High-Value Customer Visibility:** The 2024 view showed very few customers with 3+ orders (high-value loyal segment) because achieving that within a single year is rare for most product categories.
- **Strategic Blindspot:** Without seeing 2025 orders, we can't identify which 2024 single-order customers were engaged and would return soon vs. truly churned customers.

BUSINESS VISUALIZATION OVERVIEW

Repeat Purchase Depth
2024 & 2025



Total Customers: 888 | Distribution: Multi-tier loyalty segments clearly visible

- **Loyalty Transformation Revealed**: Hundreds of customers upgraded from 1 order to 2+ orders.
- **Elite Customer Segment Emerges:** High-value customers with 4-5 orders are now clearly visible.
- **Three-Tier Loyalty Model Validated:** The complete dataset reveals a natural customer segmentation - approximately 35% remain at 1 order (new/low-intent), 40% reach 2-3 orders (engaged/standard loyalty), and 25% achieve 4+ orders (power users/brand advocates).
- **239 New Customers Added Context:** The +239 new customers from 2025 aren't just volume growth - they're early in their customer journey and many still show 1 order because they're recent acquisitions.
- **Revenue Attribution Correction:** Adding 2025 orders fixed a massive revenue attribution problem - orders that were happening in 2025 can now be properly linked back to the 2024 cohort that generated them.

BUSINESS VISUALIZATION OVERVIEW

Strategic Actions Based on Combined Insights

- **Immediate Action:** Extract all customers with 4-5 orders → Create VIP tier with exclusive offers.
- **Segmentation strategy:** Apply different retention tactics to 1-order, 2-3 order, and 4+ order segments.
- **LTV recalculation:** Update customer lifetime value models to include multi-year order patterns.
- **CAC justification:** 2024 acquisition costs now show positive ROI when 2025 revenue is included.
- **Win-back campaigns:** Only target true 1-order customers after waiting period (not recent acquires).

“The combining insights transforms our understanding from "most customers only buy once" to "customers progress through loyalty tiers over time, with a valuable 25% reaching power-user status" - completely changing how we should invest in retention vs. acquisition.”

PERFORMANCE ANALYSIS & KEY INSIGHTS

E-Commerce Cohort Analysis Pipeline

Acquisition Trend: Scalability & Volume

- **Pipeline Integrity:** The integration of 2025 incremental data successfully added 1,000 records while maintaining the historical integrity of 2024. This confirms the ELT logic is robust and non-destructive.

- **Consistent Inflow:** Monthly cohort sizes remain stable across both years. This consistency indicates **a reliable acquisition engine**, providing a steady denominator for our retention calculations and ensuring that the business is not over-reliant on a single "peak" season.

Retention & Loyalty: The Velocity of Value

- **Retention Velocity (1–3 Months):** Our "Time-to-Return" analysis shows a healthy engagement cycle. The transition into 2025 reveals that early-life behavior is predictable; customers are moving toward their second purchase within the expected window, proving that our initial value proposition is strong enough to trigger repeat visits.

- **Repeat Purchase Depth (2+, 3+, 4+):** Comparing the 2024 baseline to the 2025 update reveals a significant loyalty migration. The growth in the "3+ orders" segment demonstrates that our older cohorts are not churning; they are maturing into high-frequency users, which significantly lowers our long-term Customer Acquisition Cost (CAC).

Note on Cohort Maturity

It is vital to distinguish between a performance dip and data maturity. The most recent 2025 cohorts currently show lower repeat purchase depth. This is a technical limitation, not a business failure: these customers simply haven't existed long enough to complete their natural purchase cycle. We expect these metrics to fill in and align with historical averages as more time elapses.

Conclusion:

The pipeline has moved from a proof-of-concept to a strategic asset. We have verified that our customers return across calendar years and that our data architecture can scale without friction. With this visibility, we are now positioned to shift from reactive reporting to proactive, growth-oriented decision-making for 2026.